Carlos Romero Madrid

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TRAJECTORY · HPC SYSTEMS

01

Professional Experience

BSC · Performance Team (HPC Software Eng.) 05/2025 – Present

- Ocean–atmosphere coupling in European digital twin flagships: DestinationE and ClimateDT.
- ECLand climate model development with Climate Change team.
- Deploying & monitoring long-running workflows on MN5.
- Scalable multi-node model integration across Tier-0 HPC.

BSC · Fusion Team (Research Engineer) 05/2023 – 04/2025

- Advanced Computing Hub: parallelization of nuclear fusion codes via OpenMP, MPI, PETSc, oneAPI, OpenACC on MN5.
- ALYA-MAGNET: electromagnetic modeling for fusion magnets.
- Cross-disciplinary communication with applied scientists.

CENER · Data Scientist (Analysis & Design) 05/2022 – 09/2022

- Python pipelines for real-time wind plant characterisation.
- Predictive maintenance & classification (Pandas, Scikit-Learn).
- Bridging data science analytics with operational engineering.

Univ. Seville & Fraunhofer FHR · Researcher 2018 – 2022

- Theoretical & computational plasma physics (PIC & MHD).
- 3D particle tracking in Python; Fortran magnetic equilibria.
- Lectured Quantum Physics; simulated annealing antenna design.

FOUNDATIONS · COMPUTE STACK

02

Academic Core & Toolkit

ACADEMIC EXCELLENCE (Double Degree & MSc):

- BSc Physics & BSc Mathematics (Double Degree)
  Univ. of Seville (2013–2018) · Grade: 8.35/10 · 3rd of Class
- MSc Plasma & Fusion Physics
  Univ. Aix-Marseille (2019–2020) · 15.43/20 · 1st of Class
- MSc Advanced Mathematics
  UNED (2023–Present, ABD)

TECHNICAL & HPC COMPUTING STACK:

Languages & Compilers:
Fortran, C++, Python, Bash/Shell, Git, CMake, Slurm.

Parallel & Accelerator Programming:
OpenMP, MPI, PETSc, CUDA, OpenACC, Intel oneAPI.

Tier-0 Clusters Deployed:
MareNostrum 5 (MN5), Leonardo (CINECA), LUMI (CSC).

Data Science & Machine Learning:
NumPy, Pandas, Scikit-Learn, telemetry classification.

AI-ASSISTED DEV & LANGUAGES:

- Fluid LLM pair-programming (DeepSeek, Antigravity).
- Fast multi-domain prototyping across firmware & web.

Spoken Languages:
- Spanish (Native), English (C1/Prof.), French (C1/Prof.)
- Italian (Intermediate), German (Intermediate)

Software Engineering Practices:
- Code reviews, technical documentation, agile research.

DATA ART & PHYSICAL COMPUTING

03

Creative Computing Portfolio

MNusic — Listening to & Visualizing MN5 & LUMI

- Sonifies MareNostrum 5 & LUMI telemetry into ambient sound-scapes (Web Audio API) & real-time particle wave dynamics.
- Maps power, GPU tensor spikes & cooling thermal gradients.
- Humanizes petascale algorithms for public engagement.
- Stack: HPC Telemetry, Web Audio, HTML5 Canvas, LLM dev.

Pipa-Music-Ball — Bio-Kinetic Interactive Sound

- Embedded physical sphere powered by ESP32 and 6-DoF inertial sensor (IMU accelerometer & gyroscope).
- Translates feline kinetic motion into microtonal musical phrasing.
- Non-anthropocentric tangible physical data storytelling.
- Stack: ESP32/Arduino, I2S DAC, gesture-to-tone mapping.

Expo-Curator — github.com/carrommad/expo-curator

- Automated image classification, catalog booklet publishing, and modular 2D/3D gallery wall staging application.
- Built to organize complex visual collections into spatial stories.
- Directly serves exhibition design for museums & science centers.
- Stack: JavaScript, Python, spatial layout geometry.

Lens & Motion — Photography & Adobe After Effects

- Architectural still photography (framing, geometric light) paired with procedural After Effects motion design.
- Kinetic typography & procedural particle simulation.
- Converts static scientific data plots into broadcast visual essays.
- Stack: Adobe After Effects, color grading, visual rhythm.

DAV GROUP · FERNANDO CUCCHIETTI

04

Synergy with Research Lines

1. Scientific Visualization & Storytelling:

- Turning extreme-scale physics into visceral public narratives.
- Direct match: MNusic (audible MN5) and After Effects motion.
- Democratization of science for museums, festivals & workshops.
- Domestic Data Streamers inspired tactile participatory design.

2. Industrial AI & Digital Twins (GDU-HPC):

- Direct synergy with BSC DestinationE & ClimateDT digital twins.
- Scalable model integration on MareNostrum 5 & cloud workflows.
- Predictive ML & classification pipelines from CENER wind plants.
- Applied to urban analytics and industrial multi-physics.

3. Creative Intelligence & Interactive Viz:

- Bio-kinetic hardware (Pipa-music-ball) & ESP32 sensors.
- Exhibition curation & automated sub-classification (Expo-Curator).
- Participatory spatial experiments & real-time visitor streams.
- AI-augmented rapid prototyping across disciplines.

4. Urban Data Science (AIR & REALLOCATE):

- Physics grounding: fluid dynamics, plasma transport & wind.
- AIR: Environmental air resilience as meeting & expressive space.
- REALLOCATE: Data design for climate-friendly active mobility.
- Bridging mathematical solvers with actionable public tools.